

OMNIX QUANTUM LTD

PROVISIONAL PATENT APPLICATION

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STRUCTURAL ADMISSIBILITY ENGINE FOR AUTOMATED DECISION GOVERNANCE SYSTEMS WITH PRE-PIPELINE SCHEMA VALIDATION, ENUMERATED CONSTRAINT ENCODING, AND ZERO-BYPASS BOUNDARY ENFORCEMENT

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FIELD OF THE INVENTION

The present invention relates to automated decision governance systems, and more particularly to a pre-pipeline structural admissibility layer that enforces decision admissibility at the point of input construction rather than at runtime evaluation — making structurally inadmissible decision requests unrepresentable as valid system objects. The Structural Admissibility Engine (SAE) constitutes a new architectural stratum — Layer 0 — that precedes and gates all downstream governance processing, providing a zero-bypass guarantee that no inadmissible request can enter the evaluation pipeline regardless of the operational state of any downstream component.

BACKGROUND

I. THE FUNDAMENTAL INADEQUACY OF RUNTIME INTERCEPTION

Contemporary automated decision governance systems operate within a runtime interception model: they intercept and block invalid decisions after those decisions have been formulated and submitted. This architecture has five irremediable structural deficiencies:

1.1 Bypass Through Component Failure. Every blocking component must be operational for the governance guarantee to hold. A failed, disabled, or bypassed checkpoint may allow an inadmissible decision to proceed.

1.2 Latent Invalid State Generation. The invalid decision is formulated as a system object before any checkpoint evaluates it, generating latent invalid state throughout memory, logs, and audit trails.

1.3 Governance Dependency on Execution Order. Any inversion of checkpoint execution order — through code defect, configuration error, or concurrent execution — may permit an inadmissible decision to receive an execution commitment.

1.4 Absence of a Constitutive Boundary. No architectural mechanism prevents the construction of a request for an operation categorically prohibited by jurisdiction, asset class, or ethical constraint.

1.5 No Constraint Provenance in Rejection. Runtime rejection typically produces an error code or binary BLOCKED decision, without identifying the specific structural constraint violated or its regulatory source.

II. PRIOR ART AND ITS LIMITATIONS

Prior art includes runtime checkpoint pipelines (OMNIX-PAT-2026-001), web API schema validation (JSON Schema, OpenAPI), programming language type systems, and access control frameworks (OAuth, RBAC, ABAC). No prior art combines schema-level structural validation, regulatory constraint encoding, zero-bypass guarantee, structured rejection with constraint provenance, and composable cross-domain constraint architecture as a unified pre-pipeline governance layer for automated decision systems.

SUMMARY OF THE INVENTION

Component A — Structural Constraint Schema (SCS): Declarative, machine-readable specification of all admissibility constraints in four classes: Jurisdiction-Asset, Jurisdiction-Operation, Ethical Compliance (Sharia / ESG / Sanctions), and Client-Specific constraints.

Component B — Structural Admissibility Validator (SAV): Pre-construction validator that evaluates a proposed decision request against the SCS before constructing any EvaluationRequest object. If any constraint is violated, no object is constructed.

Component C — Zero-Bypass Boundary Enforcement (ZBE): Architectural guarantee that the SAV is the sole path through which a proposed request may become an EvaluationRequest object. EvaluationRequest construction is private to the SAV.

Component D — Structured Rejection with Constraint Provenance (SRCP): Machine-readable rejection record identifying: violated constraint, constraint class, regulatory source, responsible input fields, and resolution guidance.

Component E — Composable Cross-Domain Constraint Architecture (CCCA): Constraint composition mechanism allowing constraints from multiple domains to be evaluated simultaneously via a unified Constraint Registry without runtime branching logic.

THE FOUR-LAYER GOVERNANCE ARCHITECTURE

The present invention introduces Layer 0 to the governance architecture. **Key distinction: Layers 1–3 evaluate requests that exist. Layer 0 determines what can exist.**

Layer	Name	Function
Layer 0	Structural Admissibility Engine (SAE)	Determines whether a proposed request is structurally representable as a valid system object (OMNIX-PAT-2026-001).
Layer 1	OMNIX Runtime Pipeline	Sequential multi-checkpoint pipeline (CP-0 through CP-11, TIE) evaluating signal quality, context, and compliance. (OMNIX-PAT-2026-001)
Layer 2	Trajectory Invariant Engine	Trajectory-aware evaluation against historical and projected behavioral trajectory. (OMNIX-PAT-2026-001)
Layer 3	Evidence and Receipt Layer	Post-quantum cryptographically sealed decision receipt — immutable audit record. (OMNIX-PAT-2026-001)

ZERO-BYPASS BOUNDARY ENFORCEMENT

The zero-bypass property is the defining characteristic distinguishing the SAE from prior art. EvaluationRequest is defined with a private constructor accessible exclusively via the SAV:

```
class EvaluationRequest:
    _SAV_TOKEN = object() # Private sentinel
    def __init__(self, _token, asset, operation, jurisdiction, client_id, metadata):
        if _token is not self._SAV_TOKEN:
            raise StructuralAdmissibilityViolation("Must use SAV.validate_and_construct()")
    class StructuralAdmissibilityValidator:
        def validate_and_construct(self, proposed):
            violations = self._evaluate_all_constraints(proposed)
            if violations:
                raise StructuralAdmissibilityViolation(violations)
            return
```

`EvaluationRequest(_token=EvaluationRequest._SAV_TOKEN, **proposed)`

Guarantee: silence is impossible — an inadmissible request either raises an exception or is never constructed. No code path, configuration flag, or operational condition can produce an `EvaluationRequest` without passing through the SAV.

CLAIMS

1. A computer-implemented Structural Admissibility Engine comprising: (a) a Structural Constraint Schema encoding admissibility constraints in at least two classes including jurisdiction-asset and jurisdiction-operation constraints; (b) a Structural Admissibility Validator evaluating proposed decision requests against the Schema before constructing any `EvaluationRequest` object; (c) wherein the Validator constructs the `EvaluationRequest` only if all applicable constraints are satisfied; and (d) wherein the Validator does not construct the `EvaluationRequest` if any applicable constraint is violated.
2. The SAE of claim 1, wherein the `EvaluationRequest` type comprises a private constructor accessible exclusively through the Structural Admissibility Validator, enforcing a zero-bypass property whereby no external code path can construct a valid `EvaluationRequest` object.
3. The SAE of claim 1, wherein the Structural Constraint Schema further comprises ethical compliance constraints derived from at least one of: Sharia compliance criteria, ESG screening criteria, and international sanctions lists.
4. The SAE of claim 1, wherein the Structural Constraint Schema further comprises client-specific constraints that may further restrict but may not expand the admissibility determined by jurisdiction-asset and jurisdiction-operation constraints.
5. The SAE of claim 1, wherein structural rejection returns a Structured Rejection Record comprising: constraint identifier, constraint class, regulatory source citation, identification of responsible input fields, and a resolution guidance field.
6. The SAE of claim 1, wherein the composable cross-domain constraint architecture evaluates constraints from multiple classes simultaneously via a unified Constraint Registry, extensible by constraint addition without modification to the Validator evaluation logic.
7. The SAE of claim 1, constituting Layer 0 of a four-layer governance architecture, wherein Layer 0 gates a Layer 1 runtime checkpoint pipeline accepting only `EvaluationRequest` objects — ensuring the pipeline cannot receive inadmissible inputs regardless of the operational state of any Layer 1 checkpoint.
8. The SAE of claim 1, wherein constraint evaluation operates on in-memory constraint tables with asynchronous refresh for dynamic constraint classes and atomic in-memory table updates.
9. The SAE of claim 7, wherein the four-layer architecture further comprises Layer 2, a trajectory invariant enforcement layer; and Layer 3, a post-quantum cryptographically sealed audit receipt layer.
10. A method for enforcing structural admissibility comprising: (a) receiving a proposed decision request; (b) evaluating the request against a Structural Constraint Schema before constructing any system object; (c) if admissible, constructing an `EvaluationRequest` via the SAV and forwarding to a downstream pipeline; (d) if inadmissible, returning a Structured Rejection Record and not constructing any object; and (e) enforcing zero-bypass by restricting `EvaluationRequest` construction to the SAV.

ABSTRACT

A Structural Admissibility Engine (SAE) for automated decision governance systems enforces decision admissibility at the point of input construction rather than at runtime evaluation, making structurally inadmissible decision requests unrepresentable as valid system objects. The SAE comprises a Structural Constraint Schema encoding regulatory, jurisdictional, and ethical constraints; a Structural Admissibility Validator that evaluates proposed requests before constructing any EvaluationRequest object; and a Zero-Bypass Boundary Enforcement mechanism restricting EvaluationRequest construction exclusively to the Validator. Inadmissible requests are rejected at the structural boundary with a Structured Rejection Record providing machine-readable constraint provenance. The SAE constitutes Layer 0 of a four-layer governance architecture gating a downstream runtime checkpoint pipeline (Layer 1), trajectory invariant enforcement layer (Layer 2), and cryptographic receipt layer (Layer 3). The zero-bypass property guarantees that no inadmissible request can enter the governance pipeline regardless of the operational state of any downstream component — a categorical improvement over runtime interception architectures where governance guarantees are contingent on every intercepting component being operational and correctly implemented.